```
import processing.serial.*; // add the serial library
Serial myPort; // define a serial port object to monitor


float x_com = 0;
float y_com = 0;
float W = 0;
int motorMoved = 0;
int isReady = 0;

void setup() {
  size(800, 800); // set the window size
//  println(Serial.list()); // list all available serial ports
  myPort = new Serial(this, Serial.list()[2], 115200); // define input
port
  myPort.clear(); // clear the port of any initial junk

 background(255, 255, 255);
}

void draw () {
  background(255, 255, 255);
  if (myPort.available () > 0) { // make sure port is open
    String inString = myPort.readStringUntil('\n'); // read input string

   if (inString != null) { // ignore null strings
   inString = trim(inString); // trim off any whitespace

    if (inString.equals("TOO LIGHT")){
      x_com = 0;
      y_com = 0;
      W = 0;
      motorMoved = 0;
      isReady = 1;
}

  else{
    String[] raw = splitTokens(inString, ",");
    if (raw.length == 4) {
```

```
      x_com = float(raw[0]);
      y_com = float(raw[1]);
      W = float(raw[2]);
      motorMoved = int(raw[3]);
      isReady = 1;
      }
    }
   }

 }
   int plateDim = 200*3; //200mm then times 4 to scale it up. need to use
times 4 consistently

   fill(109, 187, 247);
   rect(width/2 - plateDim/2, height/2 - plateDim/2-50, plateDim,
plateDim);

   fill(0,0,0);
   textSize(20);
   text("X Position:", 30,height-100);
   text(x_com, 125,height-100);

   text("Y Position:", 30, height-70);
   text(y_com, 125, height-70);

   text("Weight:", 30, height-40);
   text(W,100,height-40);
   text("N",150,height-40);

   String message;
   if (isReady == 0){
     message = "NOT READY";
     fill(232, 32, 62);
   }
   else{
     message = "READY";
     fill(181, 201, 115);
   }
   textSize(50);
```

```
  text(message,400,height-50);

  if (motorMoved == 1){
    fill(140, 166, 53);
  }
  else{
    fill(255,0,0);
  }
  ellipse((width/2)+(x_com*3), (height/2)-50-(y_com*3),20,20);




}
```